

Transformation Rules Patterns for RDB2RDF Views

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Transformation Rules Patterns

Class Transformation Rule

A Class Transformation Rule (CTR) associated with class C has the form:

$$\psi: C(s) \leftarrow R(r), \text{URI_}R(r, s), \delta(r)$$

where:

- ψ is the CTR identifier;
- C is a class in the vocabulary V of the target ontology;
- R is a relation in the source relational schema S ;
- $\text{URI_}R(r, s)$ is the URI constructor predicate of the RDF resource corresponding to r ;
- r is a tuple variable ranging over tuples of R ;
- s is the URI generated derived from tuple r ;
- $\delta(r)$ is an optional selection condition predicate over tuples of R .

The relation R is called the *pivot relation* of the rule, and tuples of R are called *pivot tuples*. Distinct pivot tuples must generate distinct RDF resources. A CTR example is:

$$\text{ctr-company-01: org:Company}(s) \leftarrow \text{company}(c), \text{URI_company}(c, s)$$

$$\text{ctr-company-02: foaf:Organization}(s) \leftarrow \text{company}(c), \text{URI_company}(c, s), (c.type = \text{"S/A"})$$

Object Type Transformation Rule

A Object Type Transformation Rule (OTR) has the form:

$$\psi: P(s, o) \leftarrow R_D(d), B_D[d, s], R_G(g), B_G[g, o], \phi(d, g)$$

where:

- ψ is the identifier of the OTR;
- P is an object property in V whose domain D and range G are classes in V ;
- " $R_D(d), B_D[d, s]$ " is the right-hand side of the CTR ψ_D that matches class D with pivot relation R_D ;
- " $R_G(g), B_G[g, o]$ " is the right-hand side of the CTR ψ_G that matches class G with pivot relation R_G ;
- $B_D[d, s]$ and $B_G[g, o]$ are the bodies of the transformation rules formed by the URI constructor predicate and the optional conditional selection predicate related to the pivot relations R_D and R_G , respectively;

- ϕ is a relational path from R_D to R_G ;
- $\phi(d, g)$ is valid if tuple d is connected to tuple g following path ϕ .

An OTR example is:

otr-company-2: $\text{ex:hasEmployee}(s, o) \leftarrow \text{company}(c), \text{URI_company}(c, s),$
 $\text{employee}(e), \text{URI_employee}(e, o), [\text{fk_company_employee}](s, o)$

Data Type Transformation Rule

There are two patterns for Data Type Transformation Rule (DTR): *Local-DTR* and *Path-DTR*.

Local-DTR The attributes belong to the pivot relation. Its transformation rule has the following pattern:

$$\psi: P(s, v) \leftarrow R_D(d), B_D[d, s], \text{nonNull}(d.A_1), \dots, \text{nonNull}(d.A_k), \\ \text{RDFLiteral}(d.A_1, "A_1", R_D, v_1), \dots, \text{RDFLiteral}(d.A_k, "A_k", R_D, v_k), \\ T([v_1, \dots, v_k], v)$$

where:

- ψ is the identifier of the DTR;
- P is a datatype property in V whose domain $D \in V$;
- " $R_D(r), B_D[r, s]$ " is the right-hand side for the CTR ψ_D that matches class D with pivot relation R_D ;
- $A_1, \dots, A_k$ are attributes of R_D ;
- T is an optional transformation function that transforms values of attributes $A_1, \dots, A_k$ to values of property P .

Path-DTR The attributes come from a relation associated with the pivot relation. Its transformation rule has the following pattern:

$$\psi: P(s, v) \leftarrow R_D(d), B_D[d, s], R_G(g), \text{nonNull}(g.A_1), \dots, \text{nonNull}(g.A_k), \\ \text{RDFLiteral}(g.A_1, "A_1", R_G, v_1), \dots, \text{RDFLiteral}(g.A_k, "A_k", R_G, v_k), \\ T([v_1, \dots, v_k], v), \phi(d, g)$$

where:

- ψ is the identifier of the DTR;
- P is a datatype property in V with domain $D \in V$;
- " $R_D(d), B_D[d, s]$ " is the right-hand side for the CTR ψ_D that matches class D with pivot relation R_D ;
- ϕ is a relational path from R_D to R_G ;
- $A_1, \dots, A_k$ are the attributes of R_G ;
- T is an optional transformation function that transforms values of attributes $A_1, \dots, A_k$ to values of property P .